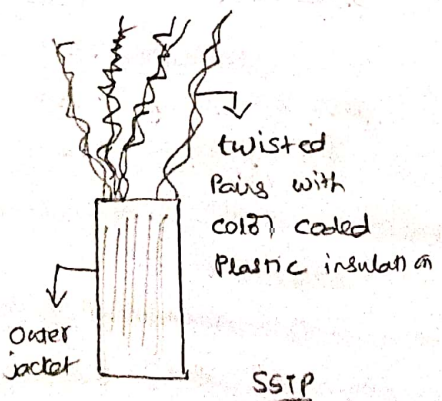
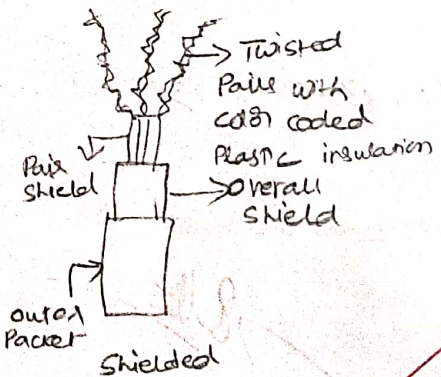
Exc: 2 STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	Category 3	10 Mbps	Adv * Cheaper in cost * Easy to install as they have a smaller overall diameter Disadv: * More Prone to (EMI) Electromagnetic interference or noise	10 Base-T Ethernet
	Category 5	upto 100 Mbps		Fast Ethernet
	Category 5e	1 Gbps		Gigabit Ethernet Fast Ethernet Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Adv: * Shielded * Faster than UTP * Less susceptible to noise and interference	Gigabit Ethernet 10G Ethernet widely used in data centres
	Category 7	10 Gbps	Disadv * Expensive * Costly installation Effort	Gigabit Ethernet 10G Ethernet (100m)

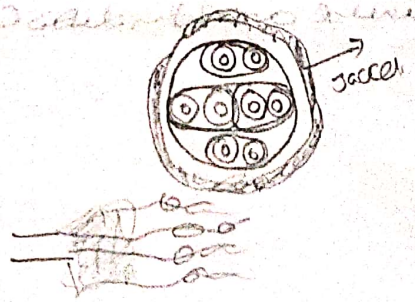
UTP

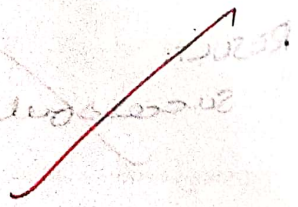


STP



SSTP

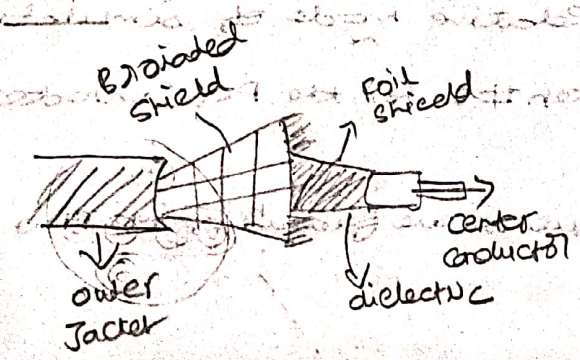




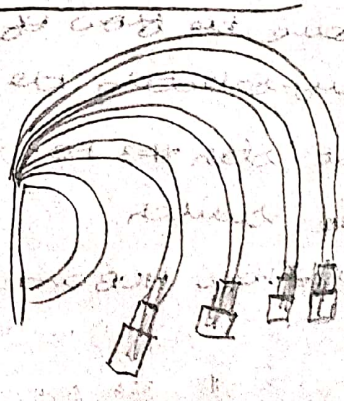
Scanned with OKEN Scanner

Coaxial Cable	RG-6 RG-59 RG-11	10-100 MbNs	Adv: * High bandwidth * Immune to interference * Less loss bandwidth * Versatile Disadvantage * Limited distance * Cost * Size is bulky	Speed of signal is 500 m Television, network High speed internet services
Fibre Optics Cable	Single Mode Multi Mode	100 GbNs	Adv: * High speed * High bandwidth * High security * Long distance Disadv: * Expensive * Requires skilled installer	Maximum distance of fibre optics cable is used 100m

Coaxial Cable



Fibre optic cable



RESULT:

Successfully analysed the study of different types of Network Codes.